

How an optical fibre guides light

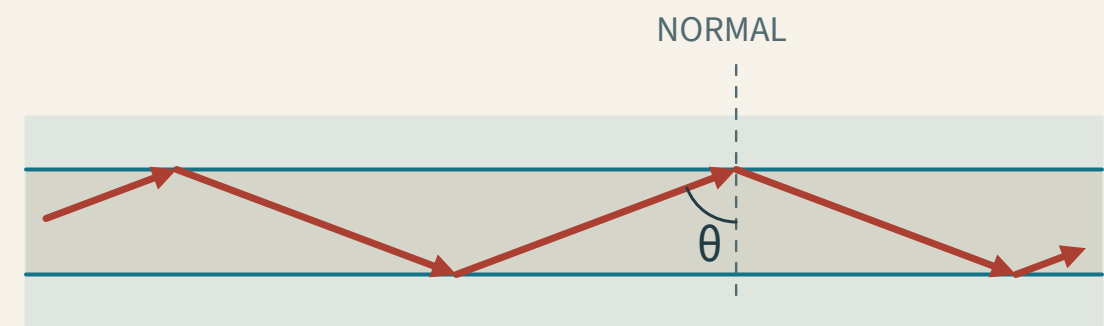
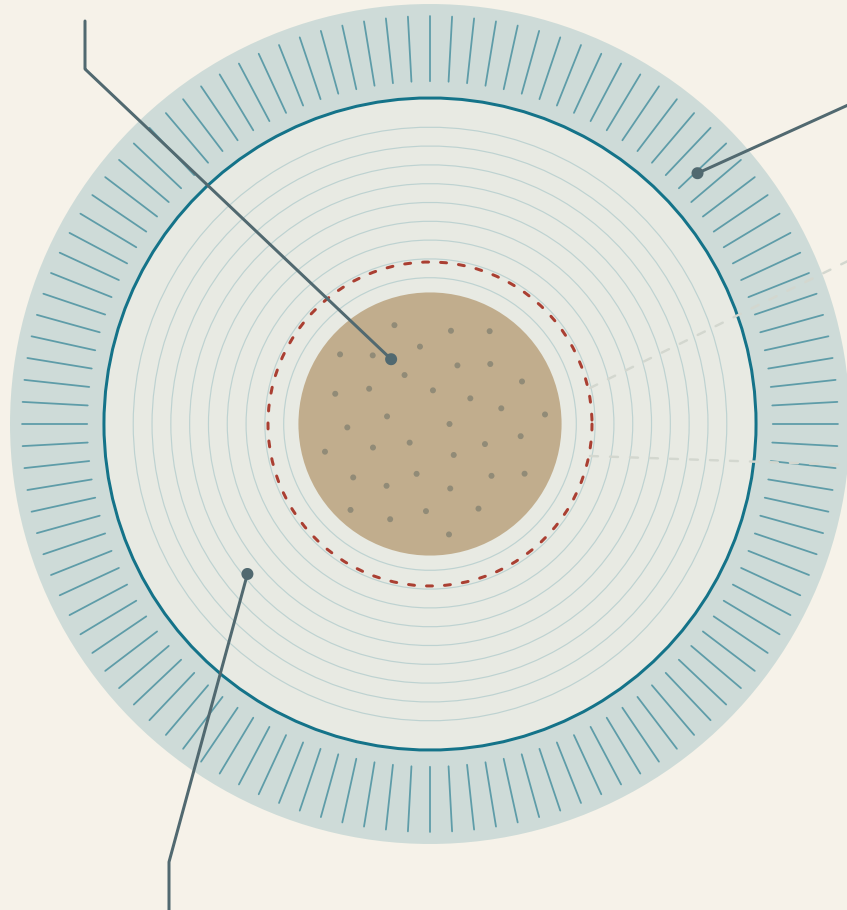
One explanation, three visual directions. A conceptual step-index fibre in the ray model.

A STRUCTURE WITH THREE ROLES

Core · n_1

Protective coating

INSIDE THE CORE



$$n_1 > n_2$$

Total internal reflection occurs at the core-cladding boundary.

Cladding · n_2

$$\theta > \theta_c$$

$$\sin \theta_c = n_2 / n_1$$

θ is measured from the normal.

The coating protects the glass; index contrast guides the light.